Kanlong Ye

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Education

- **Carnegie Mellon University** — *Aug. 2024 – May. 2026*
  M.S. in Mechanical Engineering - Research (Robotics Track) GPA: 4.0/4.0 — Pittsburgh, USA
- **Dalian University of Technology** — *Sept. 2019 – Jul. 2024*
  B.E. in Mechanical Design & Manufacturing and Their Automation (Japanese Intensive) GPA: 3.7/4.0 — Dalian, China
- **Tohoku University** — *Oct. 2022 – Aug. 2023*
  Exchange Student in Mechanical and Aerospace Engineering Department — Sendai, Japan

Publications

C=Conference, J=Journal, W=Workshop, A=Arxiv, O=Ongoing

- **[O.1]** **WindyRL: Temporal Reinforcement Learning for Vision-Based UAV Navigation in Windy Environments**. **Kanlong Ye***, Abien Cherian*, Zhefan Xu, Haoyu Shen, Hanyu Jin, and Kenji Shimada
- **[A.1]** **LV-DOT: LiDAR-visual dynamic obstacle detection and tracking for autonomous robot navigation**. Zhefan Xu*, Haoyu Shen*, Xinming Han, Hanyu Jin, **Kanlong Ye**, Kenji Shimada *arXiv:2502.20607*
- **[W.1]** **Adaptive Planning Framework for UAV-Based Surface Inspection in Partially Unknown Indoor Environments**. Hanyu Jin, Zhefan Xu, Haoyu Shen, Xinming Han, **Kanlong Ye**, Kenji Shimada *ICRA 2025 Construction Robotics Workshop*

Experience

- **CERLAB, Carnegie Mellon University** — *Aug. 2024 – Present*
  Research Assistant (Supervisor: Kenji Shimada) — Pittsburgh, USA
  - Proposed and implemented a thesis-level robot navigation system combining VLN semantic planning, wind-aware RL control, and dynamic-aware point cloud reconstruction for robust navigation in real-world environments.
  - Conducted UAV and legged-robot autonomy experiments by leveraging multiple high-impact open-source frameworks, including CERLAB UAV Autonomy, NavRL, isaac-go2-ros2, and LV-DOT.
  - Built a custom LiDAR-based UAV platform and conducted real tunnel inspection tests for Toprise Inc Japan., achieving high-resolution 3D reconstruction (accuracy < 5cm)
- **Perflection AI** — *May. 2025 – Aug. 2025*
  Software Engineer Intern — Pittsburgh, USA
  - Benchmarked state-of-the-art vision-language models (Qwen2.5-VL, Tarsier, GPT-4o, Gemini 2.5 Pro) for golf advice accuracy and curated domain-specific datasets for LoRA fine-tuning.
  - Designed and iteratively refined prompt engineering strategies (FSL, CoT), transforming generic outputs into personalized and actionable coaching feedback.
  - Incorporated key frame analysis and biomechanical metrics into the feedback pipeline, improving swing issue detection and user trust in model-generated advice.

Projects

- **Mobile Manipulation for VLN + Grasping** — *Jan. 2026 – present*
  Mobile Manipulation, VLN, Stretch 3, Embodied AI — []
  - Developing a mobile manipulation system on the Stretch 3 robot for vision–language navigation with object grasping, integrating a VLN policy based on the ApexNav framework with low-level navigation and manipulation control for long-horizon embodied tasks [Ongoing].
- **RL Algorithms** — *Jan. 2026 - present*
  Reinforcement Learning, MuJoCo, Imitation learning — []
  - Implemented a reinforcement learning pipeline covering imitation learning (BC, DAgger), policy gradient and actor-critic methods (REINFORCE, PPO, GAE), value-based RL (DQN/DDQN), and model-based RL with MPC planning, evaluated across classic control,MuJoCo, and gridworld environments [Ongoing].
- **WindyRL** — *May. 2025 - Jan.2026*
  Reinforcement Learning, UAV, Gazebo, Isaacsim — [] [][]
  - Developed Transformer-PPO architecture for wind-resilient UAV RL-control based on NavRL framework, modeling various wind fields in Gazebo/Isaac Sim and training distributed wind-aware policies that improved UAV navigation robustness.

- **Textual Attributes for Speech Understanding with LLMs** *Oct. 2025 – Dec. 2025*
  LLM, Speech Processing, Multimodal Reasoning [] [] []
  - Designed a training-free pipeline that converts speech into textual semantic, prosodic, and speaker attributes using off-the-shelf models. Achieved superior performance to Whisper-LLaMA and Qwen2-Audio on Dynamic-SUPERB core tasks, validated through ablation studies.
- **Rank-Supervised SKU Fingerprints for Low-Latency Generative Visual Search** *Oct. 2025 – Dec. 2025*
  Vision-Language Retrieval, CLIP, Diffusion, VLA [] [] []
  - Built a production-friendly SKU-per-vector fashion retrieval system on DeepFashion2, unifying image and text queries into a single embedding space. Integrated offline Stable Diffusion v1.5 img2img + LoRA multi-view augmentation to improve robustness to viewpoint variation without increasing inference cost.
- **Pittsburgh-RAG: Retrieval-Augmented Generation for QA on Pittsburgh** *Sep. 2025 - Oct. 2025*
  LLM, NLP, Information Retrieval [] []
  - Built an end-to-end RAG system from scratch, including data collection/annotation, document chunking, hybrid retrieval (BM25, FAISS dense retrieval, and fusion), and answer generation with Gemma-3, to support factual QA on Pittsburgh and CMU.
- **DDPM-AFHQ** *Jun. 2025 – Aug. 2025*
  Generative Models of Images, Diffusion Model []
  - Implemented a Denoising Diffusion Probabilistic Model from scratch and applied it to the AFHQ dataset for high-quality image generation. Conducted experiments with noise scheduling, U-Net architecture, achieved competitive FID scores.
- **Build-LLAMA2** *Aug. 2025 - Sep. 2025*
  Deep Learning, Natural Language Processing []
  - Implemented core components of the Llama2 transformer architecture from scratch, including GQA, feed-forward networks, Pre-LayerNorm, RoPE, AdamW and integrated parameter-efficient fine-tuning via LoRA, WiSE-FT and applied the model to tasks such as text continuation, zero-shot classification, and downstream fine-tuning.
- **ORB-SLAM3 on Various Physical Robots using ROS2** *Sept. 2024 - Dec.2024*
  SLAM, ROS2, Robotics [] []
  - Modernized the ORB-SLAM3 framework for compatibility with Ubuntu 20.04 and ROS2, and deployed the implementation on diverse robotic platforms (wheeled, aerial, and quadruped) to perform real-world localization and mapping tasks.
- **Optimal Control and A* Path Planning for Autonomous Vehicles** *Aug. 2024 - Dec. 2024*
  Optimal Control, LQR, A Search* [] []
  - Designed a Linear-Quadratic Regulator (LQR) for high-speed trajectory tracking and integrated an A* search algorithm for dynamic path re-planning to perform safe overtaking maneuvers, maintaining an average path tracking error of 0.76m.

Skills

- **Languages:** Chinese (Native), English (Fluent), Japanese (Fluent)
- **Programming:** C/C++, Python, MATLAB, Git, JavaScript
- **Frameworks & Libraries:** PyTorch, NumPy, OpenCV, ROS/ROS2
- **Robotics:** DIY Drone, Unitree Go2, HelloRobot Stretch 3
- **Software & Tools:** Isaac-sim, Gazebo, MuJuCo, AutoCAD, SolidWorks, Ansys, Wandb